

WINNER ENGINEERING INTERNAL TRAINING SERIES

MODULE II

Basic ACMV Knowledge & Design



Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service



CONTENT

1. Ducting Distribution System
2. Piping System
3. Air Terminals
4. Water Fittings
5. Air Side Equipment
6. Water Side Equipment

1.1 DUCT DISTRIBUTION SYSTEM

Purpose is to distribute the air from the source to the final destination point.

Things to consider:

Materials

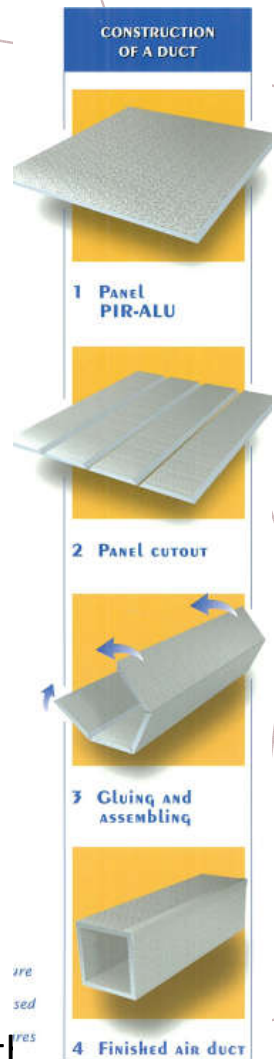
- Galvanised steel (most common usage)
- Stainless steel (kitchen duct, smoke extraction system)
- Pre-insulated duct (can be proposed due to certain advantages, eg; lightweight, time saving for installation)

Shape for steel duct

- Round duct (cost saving in terms of constructability, material, and installation)
- Square duct, more flexible for height and width ratio for site constraints.

Thickness

- Duct sheet thickness varies accordingly to the required air flow and static pressure in the duct (Refer to installation details table)



TYPE OF DUCTING SYSTEM

1. SAD – Supply Air Duct (Cold air from equipment to air terminal)
2. FAD – Fresh Air Duct (Outdoor air intake to equipment)
3. RAD – Return Air Duct (Recirculated air from the room to the equipment)
4. EAD – Exhaust Air Duct (Extracted air from indoor to outdoor)
5. PAD – Pre-cool Air Duct (Pre-cool air from equipment to equipment or air terminal)

INSULATION THICKNESS FOR DUCT IN DIFFERENT CONDITIONS

TYPE	CONDITION	INSULATION THICKNESS (MM)
SUPPLY	NO AC SPACE	50
SUPPLY, FRESH AIR	AC SPACE	25
RETURN, EXHAUST	NO AC SPACE	25
SUPPLY,RETURN,EXHAUST	EXPOSED TO WEATHER	50 / HYBRID PLASTER
SUPPLY, RETURN, EXHAUST	BELOW THE ROOF	50

1.2 DUCT SIZING

Proper duct size is important to ensure the stability of the air distribution system.

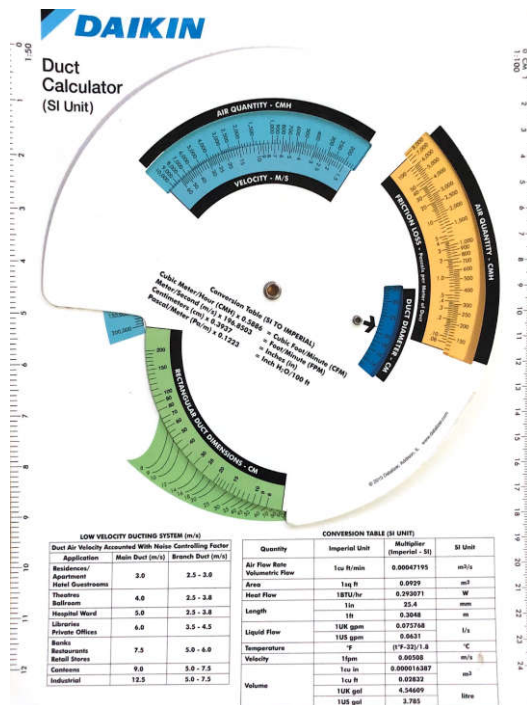
- Oversized duct resulted in material wastage, low air flow, face velocity and throwing distance from air terminal
- Undersized duct resulted in high static pressure and unwanted noise generation due to turbulence and friction.

Precon to duct design and sizing.

1. Understand the project scope of work, find out with OIC to check with Main Contractor if PID or round duct is acceptable.
2. Round duct is preferred for cost saving.

Duct sizing by using

1. Ductulator hardcopy
2. Ductulator Software



Duct Sizing																
Block	Description	Key in		Key in		Calculated										Check
		Air Flow		Duct Size		Aspect Ratio (c/d to b)	De	Area	Velocity	Velocity Pressure	Hydraulic Diameter	Absolute Roughness	Reynold Number	Friction Factor		
		CMH	L/S	W (m)	D (mm)										mm	m²
FAF SLL-1	11000	3056		700	650	1.1	730	0.428	7.1	30.75	474	0.05	319077.363	0.015	0.70	
	5500	1528		700	400	1.8	573	0.258	5.9	23.14	309	0.05	200173.5416	0.016	0.70	
	360	100		500	300	1.4	456	0.163	0.6	0.23	412	0.05	16750.11742	0.028	0.02	
	720	200		500	350	1.4	456	0.163	1.2	0.91	412	0.05	33500.23405	0.024	0.05	
	1080	300		500	350	1.4	456	0.163	1.8	2.04	412	0.05	50250.35277	0.022	0.11	
	1440	400		500	350	1.4	456	0.163	2.5	3.62	412	0.05	67000.4697	0.021	0.16	
	1800	500		500	350	1.4	456	0.163	3.1	5.65	412	0.05	83750.58712	0.020	0.27	
	2160	600		500	350	1.4	456	0.163	3.7	8.14	412	0.05	100500.7046	0.019	0.37	
	2520	700		500	350	1.4	456	0.163	4.3	11.08	412	0.05	117130.8222	0.018	0.48	
	2880	800		500	350	1.4	456	0.163	4.9	14.47	412	0.05	134000.9394	0.018	0.61	
	6450	1803		700	500	1.4	545	0.327	8.5	18.25	503	0.05	213816.9084	0.016	0.52	
	900	83		700	150	1.7	210	0.035	2.4	3.49	208	0.05	29949.52496	0.023	0.47	
	2433	676		700	300	1.4	545	0.327	2.1	3.58	208	0.05	80155.08167	0.020	0.50	
	2433	676		700	300	2.3	491	0.189	1.6	2.68	410	0.05	99591.97317	0.019	0.54	
	811	225		500	300	1.7	420	0.138	1.6	1.6	375	0.05	40500.80791	0.023	0.10	
	3920	1089		700	350	2.0	534	0.224	4.9	14.25	467	0.05	150732.2796	0.017	0.54	
	2340	650		400	250	2.4	414	0.135	4.8	14.06	353	0.05	113217.4649	0.019	0.71	
	780	217		400	250	1.6	344	0.093	2.3	3.28	308	0.05	47653.00246	0.022	0.24	
	1600	444		500	250	2.0	381	0.114	3.9	9.16	337	0.05	86376.41233	0.020	0.54	
	4700	1306		400	400	1.5	333	0.223	8.9	26.64	400	0.05	184508.6636	0.017	0.73	
	240	74		400	200	1.3	378	0.112	0.8	0.43	343	0.05	17565.27997	0.027	0.05	
	1000	278		300	400	0.8	378	0.112	2.5	3.7	343	0.05	56380.05072	0.021	0.23	
	3710	1031		200	200	1.0	219	0.038	27.4	451.05	200	0.05	363506.0115	0.017	19.29	
	1855	515		150	150	1.0	164	0.021	24.4	398.57	150	0.05	243076.7375	0.019	14.12	
	550	153		400	200	1.3	378	0.112	1.4	1.12	343	0.05	31009.0323	0.024	0.08	
	600	167		300	200	1.5	267	0.056	3.0	5.24	240	0.05	47460.89345	0.023	0.50	
	300	83		250	150	1.7	210	0.035	2.4	3.49	208	0.05	29949.52496	0.023	0.47	
	2470	688		300	200	1.5	267	0.056	7.3	22.04	240	0.05	116279.184	0.019	2.56	
2940	817		500	250	2.0	381	0.114	7.2	20.93	333	0.05	158475.1303	0.018	2.46		
4900	1361		400	400	1.5	333	0.223	4.1	22.43	400	0.05	274576.5217	0.017	0.79		
3710	1031		500	400	1.3	499	0.188	5.5	18.15	444	0.05	162020.3609	0.017	0.71		

DUCTULATOR

File Calculations Help

Calculations Method

☐ Velocity

☒ Friction Rate

☐ Duct Size

Units

☒ SI

☐ British

Inputs

Air Flow
278 Lps

Friction Rate
0.6 Pa/m

Duct Depth
200 mm

Outputs

Velocity
3.82 m/s

Duct Diameter
305 mm

Duct Width
400 mm

Reset Exit

1.3 DUCT SIZING RULE OF THUMB

Considerations

Static pressure of system, air flow, velocity, noise.

Approach 1

- Sizing based on static loss, (standard 0.6 to 1.0)Pa/m
 - indicates the amount of static losses in the system per meter duct run.
 - good practice to size main duct at 0.6 for lesser static loss.

Approach 2

- Sizing based on velocity
 - If the duct sized based on approach 1 has a **velocity of >10m/s**, to size again based on max. 10m/s.
 - Considered as secondary approach due to spaces like library which require lesser dBA ambient.

$$* \text{Velocity (m/s)} = \text{Air Flow (m}^3\text{/s)} / \text{Duct Area (m}^2\text{)}$$

2.1 PIPING SYSTEM

- To delivery as a pathway for the flow of medium (water/refrigerant) of heat transfer to occur.

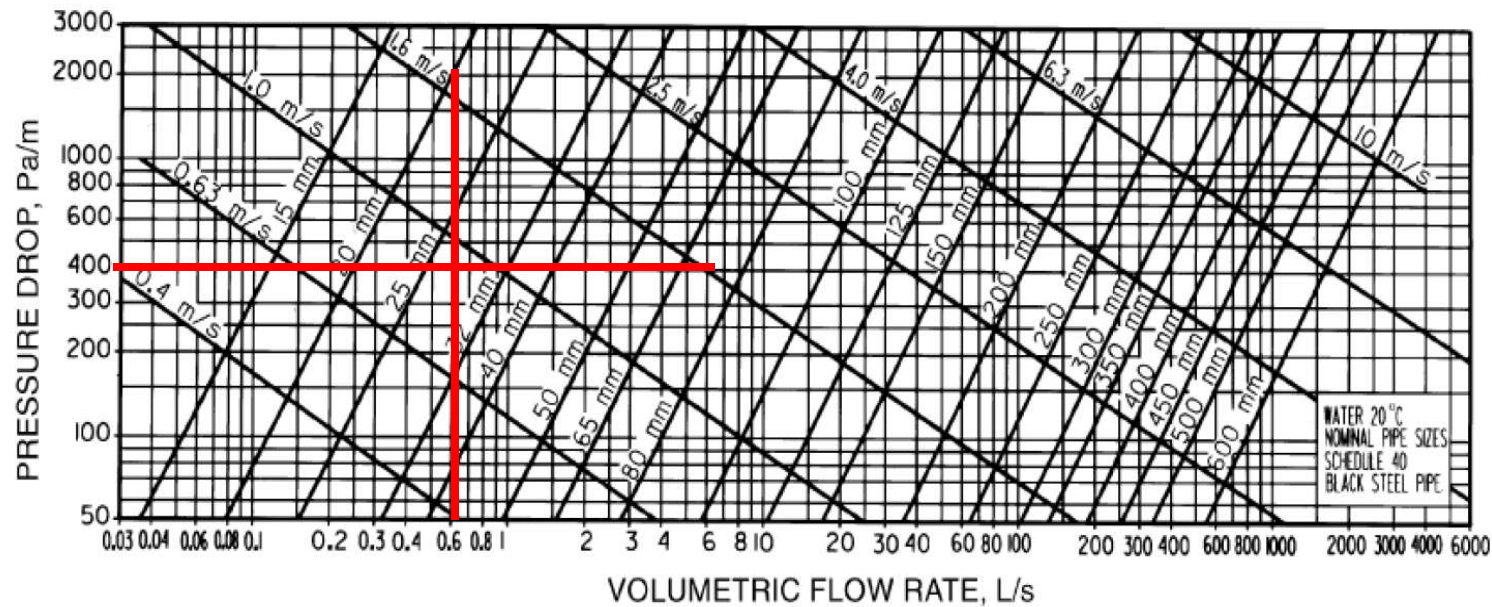
Types of Common Piping System

1. CHWS - Chilled Water Supply
2. CHWR - Chilled Water Return
3. CDP - Condensate Drain Pipe
4. REF. - Refrigerant pipe
5. *CHW Loop - Chilled water loop from chiller evaporator to air side equipment
6. *CW Loop - Condenser water loop from chiller condenser to cooling tower.

2.2 PIPE SIZING

Chilled Water Piping System

Table 1: Friction Loss for Water in Commercial Steel Pipe (Scheduled 40)



2.2 PIPE SIZING

CONDENSATE DRAIN PIPE

COOLING CAPACITY (KW)	DRAIN PIPE SIZE (MM)
BELOW 17	ø25
17.1 ~ 35	ø32
35.1 ~ 175	ø40
ABOVE 175	ø50

3.1 AIR TERMINAL

- To dissipate or extract air from the ductwork distribution system

Types of Common Air Terminal

1. SAG,SAL : Supply Air Grille, Supply Air Louvre
2. RAG : Return Air Grille
3. EAG,EAL : Exhaust Air Grille, Exhaust Air Louvre
4. LSAD,LRAD : Linear Supply Air Diffuser, Linear Return Air Diffuser

Types of Features

- Louvre : Insect/vermin proof, weather proof
- Grille : Anti-condensation, SDV,DDV,2-way,3-way,4-way
- Linear Diffuser : 2 slot,4 slot, (Depends on air flow and throwing distance requirement)

3.2 SIZING AIR TERMINAL RULE OF THUMB

- Undersized air terminal resulted in high static pressure loss, and noise generation
- Sizing is normally done based on the following common practice parameters
 1. Face velocity of maximum 2.5m/s
 2. Free area opening of 50% and 70% for louvres and grilles respectively

Approach 1

- Manual calculation

Approach 2

- Excel sheet (refer to e-cabinet)

$$\text{Velocity} = \text{Air Flow} \div \text{Area}$$
$$\text{m}^3/\text{s} = (\text{m}^3/\text{s}) \div \text{m}^2$$

3.2 SIZING AIR TERMINAL RULE OF THUMB

Approach 1

Manual calculation principle

Velocity = Air Flow $\div$ Area

Area = Air Flow $\div$ Velocity

$m^2 = (m^3/s) \div m/s$

$m^2 = (CMH \div 3600) \div m/s$

For external louvre of 50% opening, and face velocity of 2.5m/s,

Therefore, $0.5m^2 = (CMH \div 3600) \div 2.5m/s$

*By limiting the face velocity to 2.5m/s, the static pressure loss across the louvres would be limited to 50Pa or less.

3.2 SIZING AIR TERMINAL RULE OF THUMB

Approach 2

Excel Sheet

